NPTEL Online Certification Course
Applied Accelerated Artificial Architecture
Assignment 9 : Detailed Solution
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Q1. Which graph capture mechanism cannot handle dynamic control flow (such as if-else statements) because it is highly tensor shape specific?

- torch.export
- torch.jit.script
- torch.jit.trace
- torch.compile

Correct Answer: **torch.jit.trace**

Explanation: `torch.jit.trace` records operations based on a single execution pass with example inputs. Therefore, it silently ignores alternative control flow paths not taken during that specific trace, making it unsuitable for highly dynamic logic.

Q2. What is the recommended modern API for capturing graphs with dynamic shape support in PyTorch versions 2.x and later?

- torch.jit.trace
- torch.export
- torch.jit.script
- torch.cuda.graph

Correct Answer: **torch.export**

Explanation: Introduced in PyTorch 2.0, `torch.export` is the standard ahead-of-time (AOT) compilation tool designed to capture full graph structures. It reliably handles dynamic shapes and produces a standardized ExportedProgram suitable for stable deployment.

Q3. During a transformer forward pass execution on an NVIDIA GPU at batch size 1, what primary issue causes the GPU to remain idle while waiting for the next kernel call?

- Matrix multiplication latency
- CPU launch overhead
- Flash attention tiling

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- GPU memory fragmentation

Correct Answer: **CPU launch overhead**

Explanation: At a batch size of 1, the computational workload per kernel is so small that the GPU finishes executing tasks faster than the CPU can queue them up. This bottleneck leaves the GPU idle while waiting for the CPU to dispatch subsequent instructions.

Q4. What are the graph nodes represented as in an ONNX graph?

- Edge devices
- Core weights
- Tensor edges
- Operators

Correct Answer: **Operators**

Explanation: In the ONNX (Open Neural Network Exchange) computational graph model, the individual nodes represent mathematical or functional operators (such as Convolution, ReLU, or MatMul). The edges that connect these nodes represent the tensors flowing between the operations.

Q5. Why can an execution provider fallback hurt the overall performance during model serving?

- It disables the operator fusions permanently.
- It causes a reduction in model parameters.
- It introduces data transfer and synchronization overhead.
- It changes static shapes into symbolic ones.

Correct Answer: **It introduces data transfer and synchronization overhead.**

Explanation: When a hardware accelerator (like a GPU) cannot execute a specific node, the ONNX Runtime falls back to the CPU. This forces data to be actively transferred between the device memory and the host memory, creating significant latency and synchronization delays.

Q6. In which scenario is the ONNX runtime considered an ideal choice for model deployment?

- When maximizing the throughput of a specific specialized hardware target
- When deploying cross-platform models across various target environments
- When executing highly customized neural networks on restricted single-node systems
- When managing complex training loops and backpropagation tasks

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Correct Answer: **When deploying cross-platform models across various target environments**

Explanation: ONNX Runtime is inherently designed for wide interoperability and portability. It allows a single exported model to be deployed seamlessly across different operating systems, hardware architectures, and accelerators using various execution providers.

Q7. What optimization happens during the "tactic selection" phase of TensorRT's pipeline?

- It permanently fusions layers together.
- It selects the fastest kernel for an operation.
- It reduces the memory fragmentation.
- It adjusts the dynamic shape range boundaries.

Correct Answer: **It selects the fastest kernel for an operation.**

Explanation: During tactic selection, TensorRT automatically profiles multiple hardware-specific kernel implementations (tactics) for a given layer. It benchmarks these candidate kernels directly on the target GPU to choose the one that delivers the fastest execution time.

Q8. Which specific optimization technique is implemented in TensorRT-LLM to reduce memory fragmentation and improve concurrency?

- Shape specialization profiles
- Operator kernel serialization
- Paging for large KV caches
- Explicit data type fallback

Correct Answer: **Paging for large KV caches**

Explanation: TensorRT-LLM utilizes PagedAttention, which treats the Key-Value (KV) cache like virtual memory mapped into non-contiguous blocks (pages). This optimization essentially eliminates memory fragmentation and allows the system to process much larger batches concurrently.

Q9. Why is engine building in TensorRT considered a decoupled and expensive process?

- It repeatedly executes complete backpropagation steps.
- It downloads pre-optimized architectures over the network.
- It converts symbolic layouts to static arrays dynamically.
- It explores and tests multiple candidate kernels.

Correct Answer: **It explores and tests multiple candidate kernels.**

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Explanation: The engine building phase involves an exhaustive local search where TensorRT profiles hundreds of potential kernel fusions and implementations on the specific target hardware. This benchmarking process takes considerable

Q10. What does the parameter "zero point" represent in the context of quantization?

- The maximum integer value in a scale
- The margin of error after reconstruction
- The integer code representing real zero
- The spacing between representable labels

Correct Answer: The integer code representing real zero

Explanation: In asymmetric quantization, the "zero point" is the quantized integer value that maps exactly to the real-world floating-point value of 0.0. This guarantees that operations like zero-padding, which are common in neural networks, introduce no approximation errors.

Q11. What is a key difference between Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT)?

- PTQ is much more expensive to run than QAT.
- QAT simulates quantization during the training or fine-tuning stage.
- PTQ provides a higher baseline accuracy than QAT.
- QAT requires storing less metadata than PTQ.

Correct Answer: QAT simulates quantization during the training or fine-tuning stage.

Explanation: While PTQ directly converts a pre-trained model's weights without any retraining, QAT integrates fake quantization nodes into the computational graph during the training process itself. This allows the model to learn and adapt to quantization errors, resulting in higher accuracy.

Q12. Why is large language model (LLM) quantization trickier compared to other traditional deep learning models?

- LLM weights and activations can contain outlier channels.
- Traditional deep learning models do not use activation clipping.
- LLMs cannot utilize approximate second-order information.
- LLM hardware architectures lack support for matrix multiplication.

Correct Answer: LLM weights and activations can contain outlier channels.

Explanation: Large Language Models frequently exhibit extreme outlier values in specific activation channels that are crucial for accuracy. Quantizing these dramatic outliers with

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standard broad scales drastically degrades model performance without specialized scaling techniques.

Q13. Which technique utilizes approximate second-order information (usually the Hessian matrix) to compensate for layer-wise quantization errors?

- Static PTQ
- AWQ
- GPTQ
- Dynamic PTQ

Correct Answer: GPTQ

Explanation: GPTQ (Generative Pre-trained Transformer Quantization) uses an optimal brain quantization approach that relies on the inverse Hessian matrix. As it iteratively quantizes weights, it adjusts the remaining unquantized weights to explicitly compensate for the introduced error.

Q14. The Activation-error Weight Quantization (AWQ) technique primarily relies on calibration data to identify which of the following?

- Backpropagation thresholds
- Outlier hardware nodes
- Linear regression targets
- Salient activation channels

Correct Answer: Salient activation channels

Explanation: AWQ observes a small set of calibration data to determine which activation channels are the most "salient" or important to the model's overall output. It then protects the corresponding weights associated with these salient channels from aggressive quantization.

Q15. What is a key benefit of using group-wise scaling over choosing a single scale for an entire matrix?

- It avoids the need to store any metadata.
- It minimizes the quantization error.
- It completely eliminates the need for suitable kernels.
- It drastically reduces the runtime memory load.

Correct Answer: It minimizes the quantization error.

Explanation: Group-wise scaling divides a large weight matrix into smaller functional blocks (groups) and computes an independent scale and zero point for each block. This

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localized precision significantly reduces the overall quantization error compared to using a single scale for the entire tensor.

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